DSAA3073: Theories in Data Science

Week 7 Validation Sheet · AdaBoost: Algorithm, Training Certificate, and Limits

Coverage: Week 7 **Duration:** 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. Which observation, by itself, establishes the population weak-PAC guarantee taught in Week 7?

(a) A base hypothesis has weighted error 0.40 in one round on one fixed sample.
(b) Every training round in one run has weighted error below $\frac{1}{2}$.
(c) For every population distribution in the declared realizable scope and every δ, a polynomial-resource learner achieves error at most $\frac{1}{2} - \gamma$ with probability at least $1 - \delta$, for one fixed $\gamma > 0$.
(d) The final ensemble has zero error on its training sample.

Solution. (c). A weak-PAC guarantee contains population-distribution and confidence quantifiers, a fixed positive edge, and a resource statement. Options (a) and (b) concern weighted empirical errors on one fixed sample; option (d) is also empirical. None of those sample observations alone supplies the population guarantee.

Marking. *5 marks for (c) together with the population-versus-fixed-sample distinction; 0 otherwise.*

A2 [5 marks]. At the start of AdaBoost round t, what is D_t?

(a) The unknown population distribution on (X, Y).
(b) A positive normalized distribution on the indices of the fixed training sample.
(c) The vector of real ensemble scores.
(d) The unnormalized exponential factors after round t.

Solution. (b). $D_{t(i)} > 0$ and $\sum_i D_{t(i)} = 1$ on the finite training-index set. It controls the current weighted empirical base-learner call. It is not the population distribution, the score vector, or the as-yet unnormalized next weights.

Marking. *5 marks for (b), including positivity, normalization, and the training-index interpretation; 0 otherwise.*

A3 [5 marks]. A binary base hypothesis has current weighted error $\varepsilon_t = \frac{1}{2}$. Which statement matches the registered coefficient formula and its interpretation?

(a) $\alpha_t = 1$, so the hypothesis receives full weight.
(b) $\alpha_t = 0$, so the hypothesis adds no signed evidence.
(c) $\alpha_t < 0$, because the hypothesis is worse than random guessing.
(d) α_t is the observed test accuracy.

Solution. (b). Substitution into $\alpha_t = \frac{1}{2}\log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$ gives $\alpha_t = \frac{1}{2}\log 1 = 0$. A hypothesis at weighted error one half has no observed edge and contributes no signed score under this coefficient. The coefficient is not a test-performance statistic.

Marking. *5 marks for (b) and the zero-edge interpretation; 0 otherwise.*

A4 [5 marks]. Which conclusion follows from the Week 7 product theorem without an additional generalization argument?

(a) $\hat{R}_{S(\mathrm{sign}(F_T))} \le \prod_t Z_t$.
(b) Test error decreases at every boosting round.

(c) Zero training error determines every signed-score magnitude.
(d) Convexity alone proves a population error bound.

Solution. (a). The theorem upper-bounds empirical zero-one training error by the product of normalization factors. It does not imply monotone test behavior, does not recover signed-score magnitudes from their signs, and is not a population guarantee supplied by convexity.

Marking. *5 marks for (a) and the empirical-not-test scope; 0 otherwise.*

Part B: Definition in use

B1 [20 marks]. In one AdaBoost round, $D_t = \left(\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}\right)$ and h_t misclassifies exactly indices 1 and 2. Execute the complete round: find ε_t, α_t, Z_t, and D_{t+1} in exact form. Then explain precisely what this successful round establishes about the base learner, what all-round empirical condition would be needed for a fixed-edge training-rate corollary, and why neither statement is the population weak-learning guarantee.

Solution. The current weighted error is the total D_t-mass of the two mistaken indices:

$$\varepsilon_t = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

Since $0 < \varepsilon_t < \frac{1}{2}$, the registered coefficient is applicable and gives

$$\alpha_t = \frac{1}{2} \log \frac{\frac{3}{4}}{\frac{1}{4}} = \frac{1}{2} \log 3.$$

Thus a correct row is multiplied by $e^{-\alpha_t} = \frac{1}{\sqrt{3}}$ and an incorrect row by $e^{\alpha_t} = \sqrt{3}$. The normalizer is

$$Z_t = \frac{\frac{3}{4}}{\sqrt{3}} + \left(\frac{1}{4}\right)\sqrt{3} = \frac{\sqrt{3}}{2}.$$

The two mistaken rows have equal old weights, so after normalization they split the new total mistaken-group mass $\frac{1}{2}$ equally. The two correct rows retain their old ratio $1 : 2$ while splitting the other half. Therefore

$$D_{t+1} = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{6}, \frac{1}{3}\right),$$

which is positive and sums to one.

This calculation establishes only a roundwise empirical condition on the fixed sample: under this particular D_t, the returned hypothesis has error $\frac{1}{4}$ and observed edge $\frac{1}{4}$. To invoke one fixed-edge corollary through T rounds, one would need a single $\gamma > 0$ such that every relevant round satisfies $\varepsilon_s \le \frac{1}{2} - \gamma$. Even that all-round sample condition does not quantify over every population distribution, does not give confidence over fresh sample draws, and does not state polynomial resources. Those are separate requirements of the population weak-PAC guarantee.

Marking. *3 marks for $\varepsilon_t = \frac{1}{4}$; 3 for $\alpha_t = \frac{1}{2}\log 3$; 4 for the correct signed factors and $Z_t = \frac{\sqrt{3}}{2}$; 4 for the exact normalized vector and its sum check; 3 for identifying the conclusion as a current-round weighted empirical condition; 3 for the all-round fixed-edge condition and the missing population/confidence/resource quantifiers.*

Part C: Basic skills

C1 [7 marks]. Let $D_t = (0.20, 0.30, 0.50)$ and suppose only index 1 is misclassified. Compute ε_t, α_t, the three unnormalized next weights, Z_t, and D_{t+1}.

Solution. The error is $\varepsilon_t = 0.20$, hence

$$\alpha_t = \frac{1}{2} \log \frac{0.8}{0.2} = \frac{1}{2} \log 4 = \log 2.$$

The mistaken first row has factor $e^{\alpha_t} = 2$, while the two correct rows have factor $e^{-\alpha_t} = \frac{1}{2}$. The unnormalized weights are $(0.40, 0.15, 0.25)$, whose sum is $Z_t = 0.80$. Dividing by 0.80 gives

$$D_{t+1} = (0.50, 0.1875, 0.3125).$$

The entries sum to one, as a next-round distribution must.

Marking. *1 mark for $\varepsilon_t = 0.20$; 2 for $\alpha_t = \log 2$; 2 for the unnormalized vector; 1 for $Z_t = 0.80$; 1 for the normalized vector and sum check.*

C2 [8 marks]. Two valid rounds have $\varepsilon_1 = 0.20$ and $\varepsilon_2 = 0.30$. Using $e^{-0.26} \approx 0.7711$ and $e^{-0.16} \approx 0.8521$, compute: (i) the exact-factor product $\prod_t 2\sqrt{\varepsilon_{t(1-\varepsilon_t)}}$, (ii) the varying-edge exponential certificate, and (iii) the fixed-edge certificate based on the minimum observed edge. Order the three values from tightest to loosest.

Solution. The exact factors are $2\sqrt{0.20(0.80)} = 0.80$ and $2\sqrt{0.30(0.70)} = \sqrt{0.84} \approx 0.9165$. Their product is approximately 0.7332.

The round edges are $\gamma_1 = 0.30$ and $\gamma_2 = 0.20$. Therefore the varying-edge certificate is

$$\exp(-2(0.30^2 + 0.20^2)) = e^{-0.26} \approx 0.7711.$$

The common lower bound is $\gamma = 0.20$, so the fixed-edge certificate is

$$\exp(-2(0.20)^2(2)) = e^{-0.16} \approx 0.8521.$$

Hence $0.7332 < 0.7711 < 0.8521$: the exact-factor product is tightest and the fixed-edge specialization is loosest. All are empirical upper bounds, not observed errors or test-error guarantees.

Marking. *2 marks for both exact factors and product 0.7332; 2 for both edges; 2 for the varying-edge value 0.7711; 1 for the fixed-edge value 0.8521; 1 for the correct ordering and empirical-certificate interpretation.*

Part D: Construction and diagnosis

D1 [17 marks]. A programmer writes the following loop:

1. choose h_t and set ε_t equal to its **unweighted** fraction of mistakes;
2. set $\alpha_t = 1 - \varepsilon_t$;
3. set $D_{t+1}(i) = D_{t(i)} e^{-\alpha_t y_i h_{t(x_i)}}$;
4. after T rounds, output $F_{T(x)} = \sum_t \alpha_t h_{t(x)}$ as the predicted label.

Repair the loop in the mathematical order required by AdaBoost. For every repair, state the invariant or object type that the defective line violates.

Solution. The loop must begin from a positive normalized distribution, ordinarily $D_1(i) = \frac{1}{m}$. At round t, the base learner receives that already normalized D_t. Its error is not an unweighted count unless D_t happens to be uniform; it must be

$$\varepsilon_t = \sum_{i : h_{t(x_i)} \neq y_i} D_{t(i)}.$$

Under the stated Week 7 derivation, check $0 < \varepsilon_t < \frac{1}{2}$ before using

$$\alpha_t = \frac{1}{2} \log \frac{1 - \varepsilon_t}{\varepsilon_t}.$$

This is the exact minimizer of the binary roundwise exponential objective; $1 - \varepsilon_t$ has neither the correct scale nor that optimizing property.

Next form the unnormalized weights

$$\tilde{D}_{t+1}(i) = D_{t(i)} e^{-\alpha_t y_i h_{t(x_i)}}$$

and their total

$$Z_t = \sum_j D_{t(j)} e^{-\alpha_t y_j h_{t(x_j)}}.$$

Only then set $D_{t+1}(i) = \tilde{D}_{t+1}\frac{i}{Z_t}$. Omitting Z_t destroys the invariant $\sum_i D_{t+1}(i) = 1$, so the next weighted error need not even be a probability.

Finally, F_T is a real-valued ensemble score. The binary predicted label is $\operatorname{sign}\left(F_{T(x)}\right)$. Confusing these output types also discards information in the score magnitude. The repaired temporal order is therefore: normalized D_t, weighted error, applicability check, log-odds coefficient, unnormalized signed update, Z_t, normalized next distribution, and only after all rounds the score together with its sign prediction.

Marking. *3 marks for initialization and using the current normalized D_t; 3 for the weighted-error definition; 3 for the condition and correct log-odds coefficient; 4 for introducing Z_t and restoring normalization; 2 for distinguishing score from sign prediction; 2 for presenting the repairs in a valid temporal order with the violated invariants.*

D2 [18 marks]. At round 12, two AdaBoost runs on the same training sample both have signed training scores

$$(0.05, 0.12, 0.40, 0.75, 1.10).$$

Construct two different possible validation-error traces for rounds $12, 13, 14, 15$: one in which validation performance improves after round 12 and one in which it deteriorates. Then explain why both traces are compatible with the same zero training error, what the zero-one count discards, and what evidence would be needed for a defensible claim about later test behavior.

Solution. Every supplied signed training score is positive, so both runs have training error zero at round 12. For example, two logically possible validation-error traces are

$$(0.12, 0.10, 0.09, 0.08) \quad \text{and} \quad (0.12, 0.13, 0.15, 0.16).$$

The first improves after round 12 and the second deteriorates. Many other numerical traces with the same two directions are equally valid; the point is that the training statement imposes no numerical constraint on either.

Zero empirical classification error records only that each training sign is correct. It discards the magnitudes and distribution of $y_i F_{T(x_i)}$: for example, the first two supplied scores are close to zero, while the last two are much larger. Later rounds can move all of these real scores even while the zero-one count remains zero. They can also behave differently on untouched examples, particularly with noise, persistent hard cases, or an insufficient base class.

Therefore neither "test error must keep falling" nor "one must stop as soon as training error reaches zero" follows. A defensible empirical statement needs an honestly untouched validation or test evaluation and may report only the rounds actually evaluated without feeding the final test outcomes back into model choice. A population conclusion would additionally require a separate generalization theorem with its sampling and complexity conditions; the Week 7 training-product theorem is not such a result.

Marking. *4 marks for two explicit traces with opposite post-round-12 directions; 3 for verifying zero training error from positivity; 4 for identifying lost signed-score magnitude/distribution information; 3 for explaining why later rounds and validation behavior remain unconstrained; 4 for an untouched evaluation requirement and rejection of an automatic population/test guarantee.*

Part E: Proof and forensic derivation

E1 [25 marks]. An AdaBoost run on four training examples records $D_1 = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$. In round 1 only index 1 is wrong and $\alpha_1 = \frac{1}{2}\log 3$; the log records $Z_1 = \frac{\sqrt{3}}{2}$ and $D_2 = \left(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right)$. In round 2 only index 2 is wrong and $\alpha_2 = \frac{1}{2}\log 5$. The file records $|Z|_2 = 1$ and labels

$$W_3 = \left(\frac{1}{2\sqrt{5}}, \frac{\sqrt{5}}{6}, \frac{1}{6\sqrt{5}}, \frac{1}{6\sqrt{5}}\right)$$

as the next "distribution." The signed predictions are

$$(y_i h_1(x_i))_i = (-1,+1,+1,+1), \quad (y_i h_2(x_i))_i = (+1,-1,+1,+1).$$

Write one coherent forensic memo deciding which recorded state is corrupt, recovering the mathematically required state and missing quantity, and issuing justified verdicts on the run's claimed exact equalities, empirical training certificate, and the legality of starting a third round directly from the stored vector. Your memo must distinguish a numerical coincidence from a theorem whose conditions are actually satisfied.

Solution. The decisive consistency check is whether the object labeled a distribution has total mass one. Here

$$\sum_i W_3(i) = \frac{1}{2\sqrt{5}} + \frac{\sqrt{5}}{6} + \frac{2}{6\sqrt{5}} = \frac{\sqrt{5}}{3} \neq 1.$$

Thus W_3 is exactly the **unnormalized** round-2 update, and the record $|Z|_2 = 1$ is corrupt. The required factor is

$$Z_2 = \sum_i W_3(i) = \frac{\sqrt{5}}{3} = 2\sqrt{\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)},$$

because the round-2 weighted error under the valid D_2 is $\varepsilon_2 = \frac{1}{6}$. Dividing by this recovered factor gives

$$D_3 = \left(\frac{3}{10}, \frac{1}{2}, \frac{1}{10}, \frac{1}{10}\right).$$

We next audit the exact score identity independently of the bad label. The two signed score contributions imply exponential factors

$$\left(e^{-y_i F_2(x_i)}\right)_i = \left(\sqrt{\frac{3}{5}}, \sqrt{\frac{5}{3}}, \frac{1}{\sqrt{15}}, \frac{1}{\sqrt{15}}\right).$$

Their empirical average is

$$\frac{1}{4}\left(\frac{3}{\sqrt{15}} + \frac{5}{\sqrt{15}} + \frac{1}{\sqrt{15}} + \frac{1}{\sqrt{15}}\right) = \frac{5}{2\sqrt{15}} = \frac{\sqrt{15}}{6}.$$

The **corrected** factor product is

$$Z_1 Z_2 = \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{5}}{3}\right) = \frac{\sqrt{15}}{6},$$

so the exact telescoping equality is restored after recovering the missing normalizer. The logged equality using $Z_1|Z|_2 = \frac{\sqrt{3}}{2}$ is false. If it happened merely to upper-bound the observed error, that would be a numerical coincidence, not the product identity.

The ensemble signs can also be checked. Since $\alpha_2 > \alpha_1 > 0$, index 1 has signed score $-\alpha_1 + \alpha_2 > 0$, index 2 has $\alpha_1 - \alpha_2 < 0$, and indices 3 and 4 have positive signed scores. Hence the empirical training error is $\frac{1}{4}$. The pointwise inequality gives

$$\frac{1}{4} \leq \frac{1}{4}\sum_i e^{-y_i F_2(x_i)} = Z_1 Z_2 = \frac{\sqrt{15}}{6}.$$

This is a valid empirical product certificate because both coefficients were chosen from errors measured under the valid normalized pre-round distributions D_1 and D_2. The varying edges are $\gamma_1 = \frac{1}{4}$ and $\gamma_2 = \frac{1}{3}$, so the further valid certificate is

$$\frac{1}{4} \leq \exp\left(-2\left(\left(\frac{1}{4}\right)^2 + \left(\frac{1}{3}\right)^2\right)\right) = \exp\left(-\frac{25}{72}\right).$$

The implementation failure occurred after the second score contribution was formed, so it does not retroactively invalidate F_2 or those two-round inequalities once Z_2 is reconstructed.

It is nevertheless illegal to start round 3 from W_3 as stored. A sum of its weights over a mistake set is not a probability and therefore is not the registered ε_3; an α_3 computed from that raw mass is not licensed by the AdaBoost derivation. Normalizing to the recovered D_3 must occur before the next base-learner call and error measurement. Finally, none of the repaired identities is a test-error or population guarantee: all use the fixed training sample.

Marking. *4 marks for detecting non-unit mass and locating the corrupt round-2 normalization record; 4 for recovering $Z_2 = \frac{\sqrt{5}}{3}$ and $D_3 = (\frac{3}{10}, \frac{1}{2}, \frac{1}{10}, \frac{1}{10})$; 5 for independently reconstructing the exponential average and showing equality with corrected $Z_1 Z_2$ but not with the logged product; 4 for deriving the ensemble sign pattern and training error $\frac{1}{4}$; 4 for validating the repaired product and varying-edge certificates without turning them into test guarantees; 4 for explaining why a third-round error/coefficient computed from raw W_3 is invalid and normalization must precede it. Preserve follow-through credit after one arithmetic slip when the invariant and logical verdicts remain consistent.*

Part F: Synthesis and limitations

F1 [15 marks]. A run produces three pieces of evidence:

1. the stored next-round weights sum to 0.92;
2. differentiating the declared one-dimensional exponential objective gives the logged α_t as its unique minimizer;
3. all training signs are correct, but several signed scores move closer to zero in later rounds.

For each item, choose the useful Week 7 view—algorithmic mechanics, roundwise optimization, or signed-score evolution—and state the strongest conclusion it supports. Then assess the claim: "Together these facts prove that the next round will reduce test error."

Solution. The first item belongs to the **algorithmic view**. A next-round example distribution must sum to one, so total mass 0.92 diagnoses a missing or incorrect normalization. It supports an implementation repair: recompute the unnormalized mass and divide by its positive total before another learner call.

The second item belongs to the **roundwise optimization view**. Under the declared binary setup and $0 < \varepsilon_t < \frac{1}{2}$, derivative zero together with a positive second derivative shows that the logged coefficient is the unique minimizer of that empirical one-dimensional exponential objective. This is an objective statement on observed weighted data, not a generalization theorem.

The third item belongs to the **signed-score-evolution view**. Zero training error records only positive signs; movement toward zero reveals information hidden by that count and may signal increased sensitivity for those examples. It does not by itself determine future validation or test behavior.

The combined test-error claim is unsupported. The first item is actually a bug, while the other two concern empirical optimization and training-score evolution. Sharing AdaBoost quantities does not make the three conclusions logically interchangeable. A test claim requires honestly untouched data or a separate population generalization result with its assumptions. Therefore no Week 7 view, separately or together here, proves that the next round will reduce test error.

Marking. *3 marks for the algorithmic view and normalization diagnosis; 3 for the optimization view and unique empirical minimizer conclusion; 3 for the signed-score view and what zero-one error hides; 4 for rejecting the test-error inference with the empirical-versus-population distinction; 2 for explaining that shared state does not make the three guarantees interchangeable.*